

Pilot Assistance Requirement Extraction and Model Rationale

1. Purpose

This document explains how the initial ODME model for the Pilot Assistance use case was derived from the source use case documentation DOT/FAA/TC-21/48.

The objective is to provide traceability between:

1. source use case information
2. interpreted modelling requirement
3. ODME modelling element
4. implementation status
5. planned refinement.

The initial model is treated as a first implementation iteration. Some elements are directly supported by explicit operating limits or system descriptions, while others are modelling abstractions introduced to represent related source concepts in ODME.

2. Derivation Method

The use case was analysed by identifying five types of information:

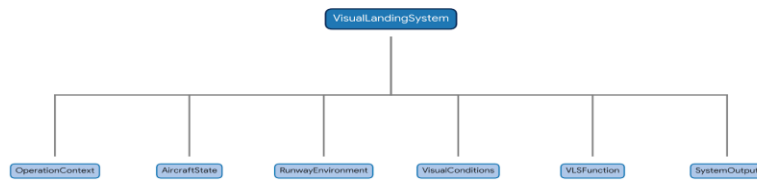
1. **Operational context**, for example Pilot Assistance and flight phase.
2. **Operational entities**, for example aircraft state, runway environment, and visual conditions.
3. **Parameters and limits**, for example distance, altitude, visibility, pitch, and roll.
4. **System functions**, for example runway detection, tracking, selection, and guidance computation.
5. **System outputs**, for example runway detection status, deviations, uncertainty, validity, and health information.

These extracted concepts were then mapped to ODME constructs:

Source concept	ODME construct
Operational object or context	Entity
Alternative operational choice	Specialization
Measurable property or state	Variable
System action or function	Behaviour
Concrete operational configuration	Pruned scenario

3. Implemented Model Structure

The initial domain model contains the following principal structure:

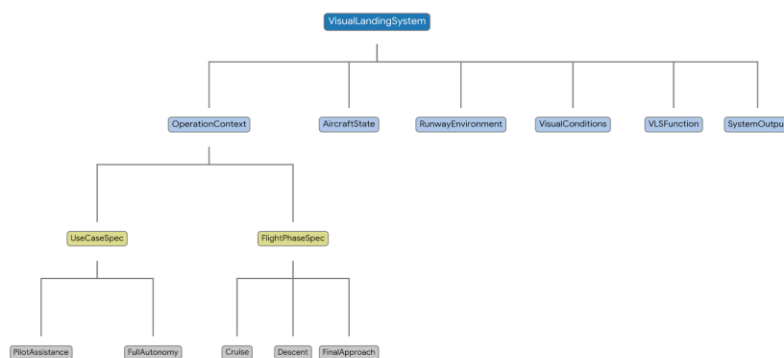


The model also contains two specialization groups under OperationContext:

```
UseCaseSpec
+-- PilotAssistance
+-- FullAutonomy
```

and:

```
FlightPhaseSpec
+-- Cruise
+-- Descent
+-- FinalApproach
```



4. Entity Rationale

4.1 OperationContext

Purpose:

Represents the operational configuration under which the Visual Landing System is being evaluated.

Reason for inclusion:

The source material distinguishes operational use cases and describes behaviour across different phases of flight. Therefore, operational mode and flight phase were represented separately from aircraft and environmental parameters.

Implemented specializations:

- UseCaseSpec
- FlightPhaseSpec

4.2 AircraftState

Purpose:

Represents aircraft position, approach geometry, and attitude parameters relevant to the operating limits of the Visual Landing System.

Implemented variables:

Variable	Range used in model	First executed scenario value
DistanceToRunwayNM	0.3 to 2.0 NM	1.5
AltitudeAGLft	100 to 2300 ft	500
GlideSlopeAngleDeg	2.5 to 8.0 degrees	5.0
LateralDeviationDeg	0 to 5 degrees	2.0
AzimuthAngleDeg	0 to 30 degrees	6.0
PitchAngleDeg	0 to 20 degrees	7.0
RollAngleDeg	0 to 30 degrees	8.5

Important clarification:

The ranges represent operating limits derived from the use case material. The executed values, such as `DistanceToRunwayNM = 1.5` and `AltitudeAGLft = 500`, were selected as valid test inputs within those ranges.

4.3 RunwayEnvironment

Purpose:

Represents runway-related assumptions and conditions relevant to system operation.

Implemented variables:

Variable	Value used in executed scenario
RunwaySurface	HardSurface
RunwayMarking	Marked
RunwayDatabaseAvailable	true
RunwaySelectedByPilot	true
RunwayIdentifierKnown	true
ParallelRunwayPresent	true

Current assessment:

The initial model represents the principal runway assumptions needed for the first workflow execution. Further refinement may include runway activity status and more detailed database attributes if they are required by future test scenarios.

4.4 VisualConditions

Purpose:

Represents environmental and visibility conditions relevant to the operational use case.

Implemented variables:

Variable	Model boundary or category	Executed value
VisibilityKM	minimum 4 km	25 km
SunElevationDeg	minimum 5 degrees	50 degrees
TimeOfDay	categorical	Daytime
WeatherCondition	categorical	VMC

Again, 25 km visibility and 50 degrees sun elevation were selected test values within the valid operational region.

4.5 VLSFunction

Purpose:

Represents system-level functions identified from the operational workflow and system description.

Implemented behaviours:

- DetectRunway
- TrackRunway
- SelectRunway
- EstimateRelativePose
- ComputeGuidanceDeviation
- ReportUncertainty
- MonitorSystemHealth
- DisplayGuidanceToPilot

These behaviours provide an initial functional abstraction of the Pilot Assistance workflow.

4.6 SystemOutput

Purpose:

Represents observable output states generated or communicated by the system.

Executed values:

Variable	Value
RunwayDetected	true
SelectedRunwayIdentifier	RWY_XX
RelativePositionAvailable	true

Variable	Value
HorizontalDeviationAvailable	false
VerticalDeviationAvailable	true
UncertaintyAvailable	false
ValidityFlag	true
SystemHealthOK	true

These values were used to test scenario parameter persistence and XML generation.

5. Requirement to Model Mapping

Requirement ID	Derived requirement	ODME implementation	Status
PA-REQ-001	Represent Pilot Assistance operational mode	PilotAssistance specialization	Implemented
PA-REQ-002	Represent alternative operational use case	FullAutonomy specialization	Implemented as alternative
PA-REQ-003	Represent flight phase context	FlightPhaseSpec	Implemented
PA-REQ-004	Represent aircraft operating state	AircraftState	Implemented
PA-REQ-005	Represent runway operating conditions	RunwayEnvironment	Implemented
PA-REQ-006	Represent visual operating conditions	VisualConditions	Implemented
PA-REQ-007	Represent runway detection	DetectRunway	Implemented
PA-REQ-008	Represent runway tracking	TrackRunway	Implemented
PA-REQ-009	Represent pilot runway selection	SelectRunway	Implemented
PA-REQ-010	Represent relative pose estimation	EstimateRelativePose	Implemented
PA-REQ-011	Represent guidance deviation computation	ComputeGuidanceDeviation	Implemented
PA-REQ-012	Represent uncertainty information	ReportUncertainty	Implemented
PA-REQ-013	Represent health monitoring	MonitorSystemHealth	Implemented
PA-REQ-014	Represent pilot-facing guidance	DisplayGuidanceToPilot	Implemented
PA-REQ-015	Represent operating limits	variable ranges	Implemented
PA-REQ-016	Produce scenario-specific representation	pruning and scenario XML	Verified

6. Current Limitations

The current model is an initial implementation intended to exercise the complete ODME workflow.

The following refinements remain possible:

- additional flight phase alternatives where justified,
- explicit runway database matching state,
- runway activity state,
- detailed monitoring behaviours,
- phase-specific outputs,
- negative scenarios based on failure conditions,
- explicit constraints between variables.

These are future refinements, not completed work.